Decreasing IL-6 Secretion in Human Leukemia Cell Line THP-1 by Reduction of Inflammation through Inhibition of the JAK/STAT Pathway Using Napabucasin

Cellular and Molecular Biology (CELL)

Background

Systemic Juvenile Idiopathic Arthritis and the JAK-STAT Pathway

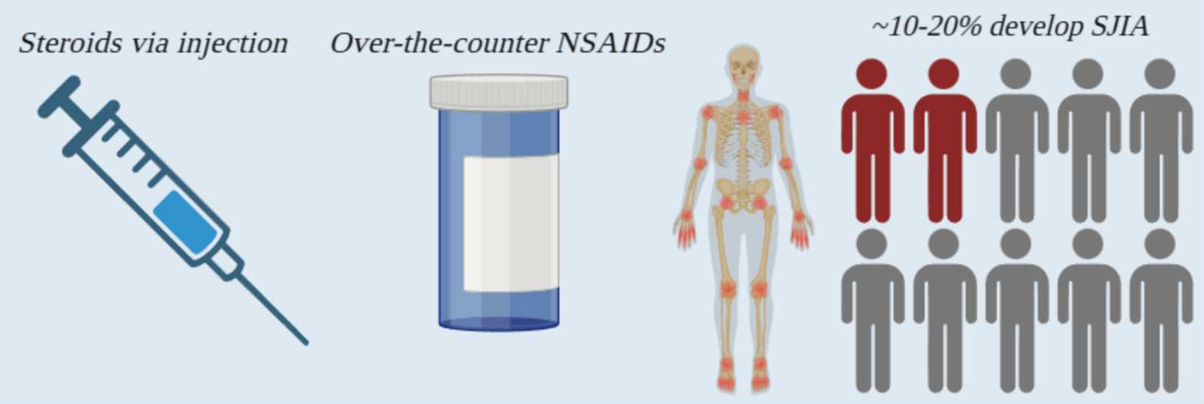
Systemic Juvenile Idiopathic Arthritis

About 3 million people worldwide have Juvenile Idiopathic Arthritis (JIA), with 300 thousand living in the United States

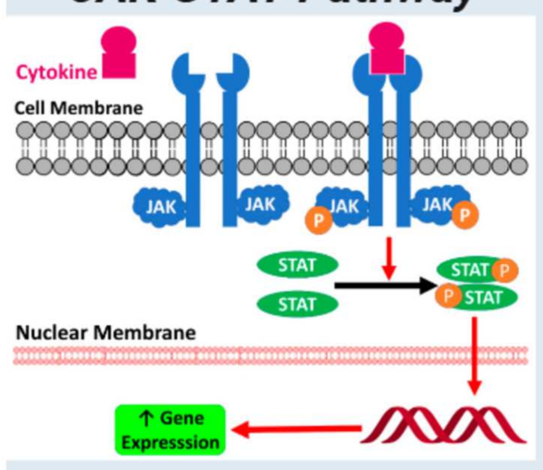
10-20% of JIA patients have Systemic Juvenile Idiopathic Arthritis (SJIA) a rare and serious subtype

Current Treatment Methods

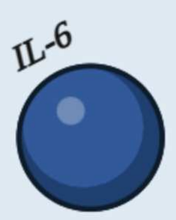
Methods include using steroids, which can have negative side effects, and NSAIDs, which are often ineffective



JAK-STAT Pathway



Inflammation plays a large role in the disease and its symptoms, one perpetrator of this inflammation is the JAK-STAT pathway. It will be targeted during research.



IL-6 is a cytokine linked with inflammation in SJIA. IL-6 expression will be measured to model inflammation levels.

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Figure 1: Image showing some brief background for this project. Links the main item looked at in this research (IL-6) to an autoimmune disease.

Purpose

Interleukin-6 (IL-6) is an **inflammatory cytokine** released by cells at the site of inflammation. The cytokine plays an **extremely vital role** in the immune response of the body, and thus helps dictate the inflammation levels of the response. When the inflammation levels decrease, **so do the levels of IL-6.** However, **this correlation is not direct**. The decrease in IL-6 levels is a **byproduct** of the decrease in inflammation. The treatments used for SJIA **do not directly target IL-6.** Thus, the experiment aims to **inhibit the Janus kinase/signal transduction and activator of transcription** (JAK/STAT) pathway of cells to **reduce inflammation**, thereby lowering the overall IL-6 expression (Zou et al., 2020).

Hypothesis/Variables/Controls

Hypothesis: If human leukemia THP-1 cells are treated with napabucasin (a natural napthoquinone), then the IL-6 secretion of the THP-1 cells will decrease as the JAK/STAT pathway will be inhibited. Napabucasin has been shown to inhibit the JAK/STAT pathway, but the effect on IL-6 secretion of cells remains to be seen (Li et al., 2020).

Independent Variable: The concentration of napabucasin that the cells were treated with. The three concentrations tested are are 0.1µM, 0.5µM, and 1µM.

Dependent Variable: IL-6 expression.

Controls: The **negative control** is a subculture of cells with no strain or treatment applied to them. This will provide a baseline. A **positive control** will be utilized through the use of lipopolysaccharides (LPS) to stimulate inflammation.

Methodology

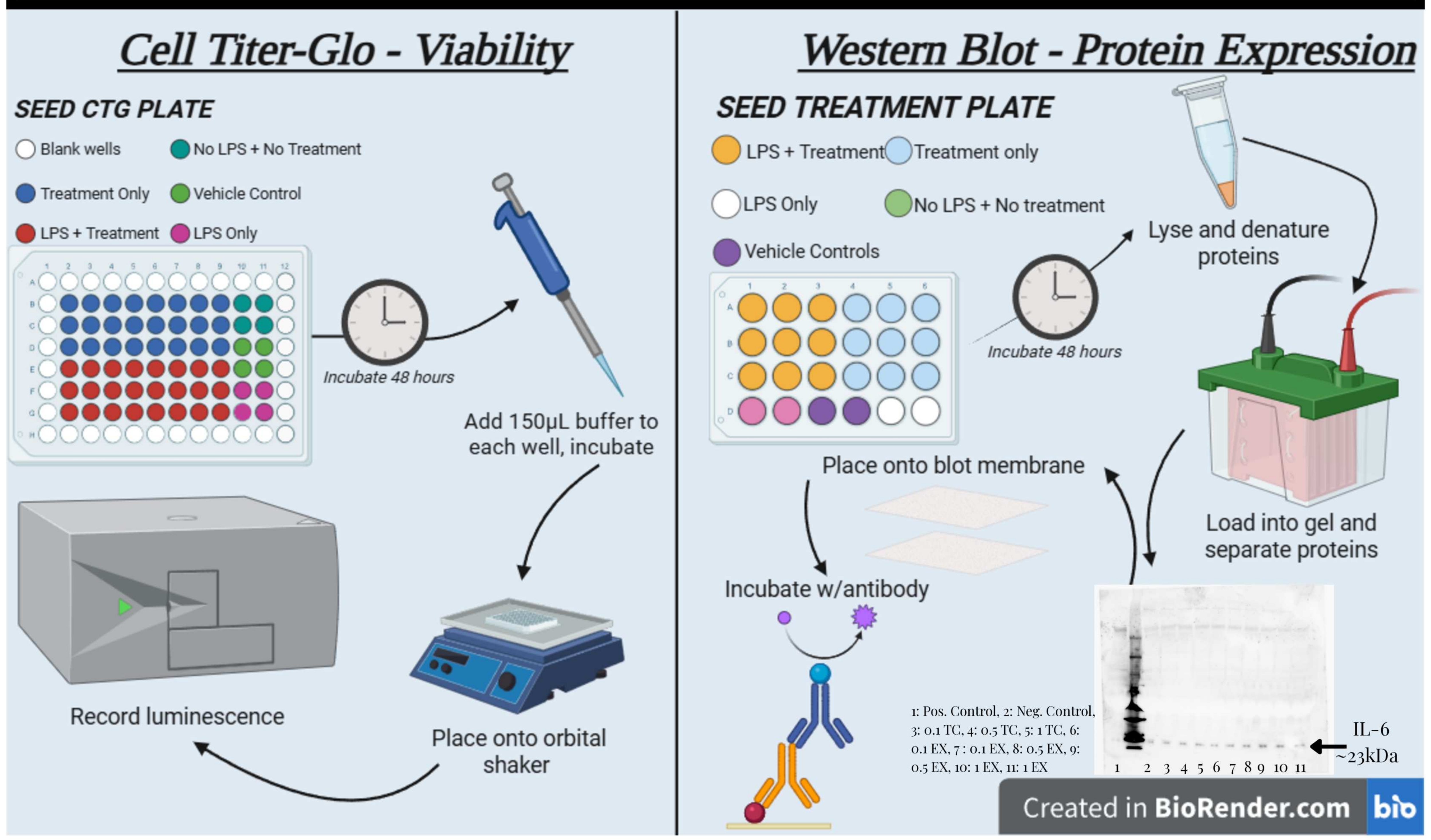

Figure 2: Diagram depicting the different methods used for this research project.

Results

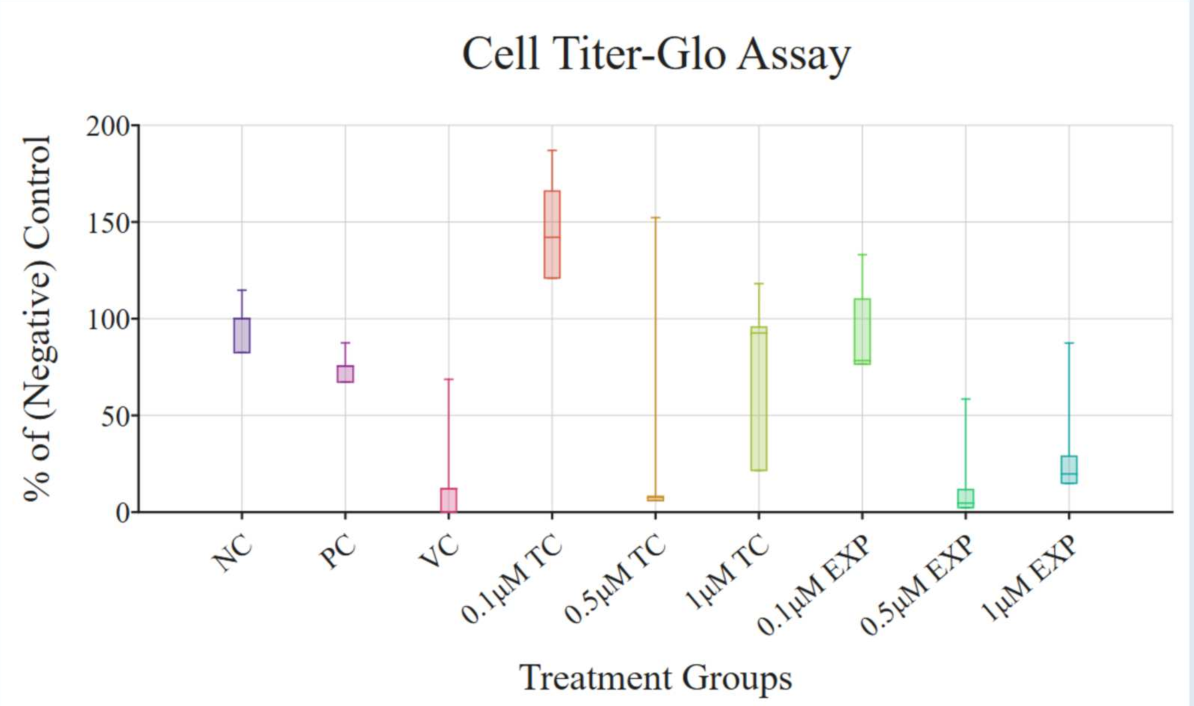

Figure 3: IL-6 Expression as percent of control of various groups.
Key: µM - Micromolar, NC - Negative Control, VC - Vehicle Control, PC - Positive Control, TC - Treatment Control, EX - Experimental

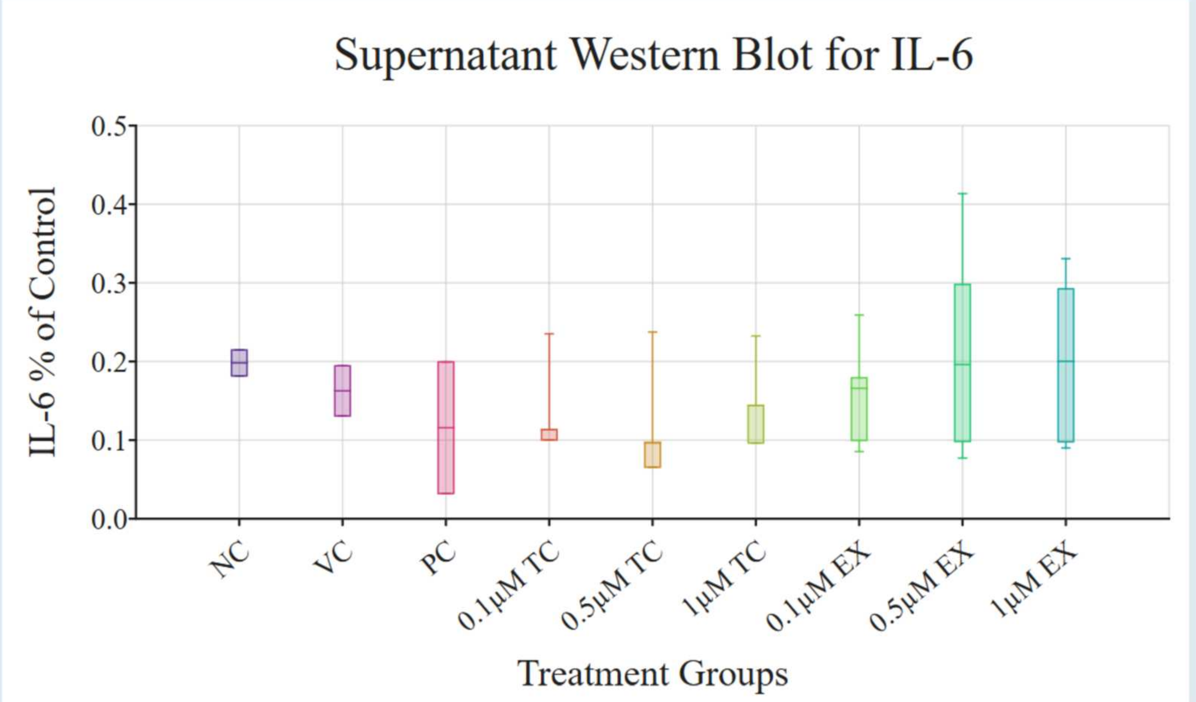

Figure 4: Viability as a percent of negative control of various groups.
Key: NC - Negative Control, VC - Vehicle Control, PC - Positive Control, TC - Treatment Control, EX - Experimental

ANOVA Results (Cell Titer-Glo)

Significant Pairs	P-value	ANOVA p-value
0.1µM TC - VC	<0.01	<0.01 (A p-value of <0.01 provides strong evidence that groups were found to be different)
0.5µM TC - 0.1µM TC	<0.01	
0.5µM EXP - 0.1µM TC	<0.01	
1µM EXP - 0.1µM TC	<0.01	

Table 1: Statistical results of an ANOVA test on cell titer-glo data.

ANOVA Results (Western Blot)

ANOVA P-Value
0.88

Table 2: Statistical results of an ANOVA test on western blot data.

Discussion

Examining **Table 2**, it can be seen that signifcant results were found for several groups. However, this is not favorable. The cell titer-glo assay measures viability, so **significant differences mean that one treatment might be less viable than another**, invalidating/skewing later results with more intensive assays like western blot. **The p-value of <0.01 for the ANOVA test signal that significant results were found**, and the specific groups found to be different are listed in **Table 1**. Looking at the **Figure 3**, these differences can be seen visually. The **clearest outliers were the vehicle control trials & the 0.5µM treatment control trials.** Half of the vehicle control data was blundered as too much vehicle was added, **killing the cells** and resulting in **zero viability**. As for the 0.5µM trial, that is more of a mystery. There is **no clear reason to point to**. The treatment at this concentration may have **degraded**.

Despite the results of the cell titer-glo assay, western blot analysis on the same treatment groups was performed. Referencing **Table 2, an ANOVA p-value of 0.88 was generated.** This suggests that there were no significant differences between any of the groups. This result **contradicts the proposed hypothesis** that an increasing concentration of napabucasin would reduce IL-6 expression.

Future Work

The data seen might have had **a major fallacy**, as the treatment solution potentially **degraded**. The results that the 0.5µM treatment group showed in the cell titer-glo assay suggest that **something went wrong**, so if this project were to be repeated, this would be **remedied**. Additionally, another go of this project would benefit from more **consistent trials**. The data provided in these trials show much **variability**, particularly in the cell titer-glo results.

If significant western blot results were generated, napabucasin could be used as a novel, therapeutic treatment of the **overexpression of IL-6** in the body by inhibiting the JAK/STAT pathway.

Other STAT3 inhibitors other than napabucasin would be something to explore if another year of research was allotted. If similar results to these were found, it would **completely dismantle the proposed hypothesis**. **Other cytokines** (IL-1β, IL-18) may be worth looking in to as well.

References

Banerjee, S., Biehl, A., Gadina, M., Hasni, S., & Schwartz, D. M. (2017). JAK-STAT Signaling as a Target for Inflammatory and Autoimmune Diseases: Current and Future Prospects. Drugs, 77(5), 521–546. https://doi.org/10.1007/s40265-017-0701-9

Figures 1 and 2 have elements created in Biorender, 2026
Figures 3 and 4 have elements created in DataClassroom, 2026

Li, X., Wei, Y., & Wei, X. (2020). Napabucasin, a novel inhibitor of STAT3, inhibits growth and synergises with doxorubicin in diffuse large B-cell lymphoma. Cancer Letters, 491, 146–161. https://doi.org/10.1016/j.canlet.2020.07.032

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Zou, S., Tong, Q., Liu, B., Huang, W., Tian, Y., & Fu, X. (2020). Targeting STAT3 in Cancer Immunotherapy. Molecular Cancer, 19(1), 145. https://doi.org/10.1186/s12943-020-01258-7